

A Personalized Video Game Recommender

Cindy Shih
Stats 418 Final Project
2026.06.02



Project Overview & Motivation

Overview

“What video game do I want to play right now?”

- The app turns a player’s current mood and situation into a ranked list of game recommendations.

Motivation

- Most recommenders emphasize rating, popularity, or genre → missing context like mood, time, or group size.
- This recommender focuses on practical inputs: play vibe, available time, and whether you are playing solo or with friends.

Data Collection & Preprocessing

Data Source

- RAWG Video Games Database API
- Fields collected: title, genres, tags, platforms, ratings, playtime, and images

Preprocessing

- Parsed nested list columns
- Removed duplicate entries
- Standardized platforms into PC, PlayStation, Xbox & Switch
- Handled missing playtime values



Model Development & Evaluation

Model Logic

Weighted scoring model

The scoring weights are:

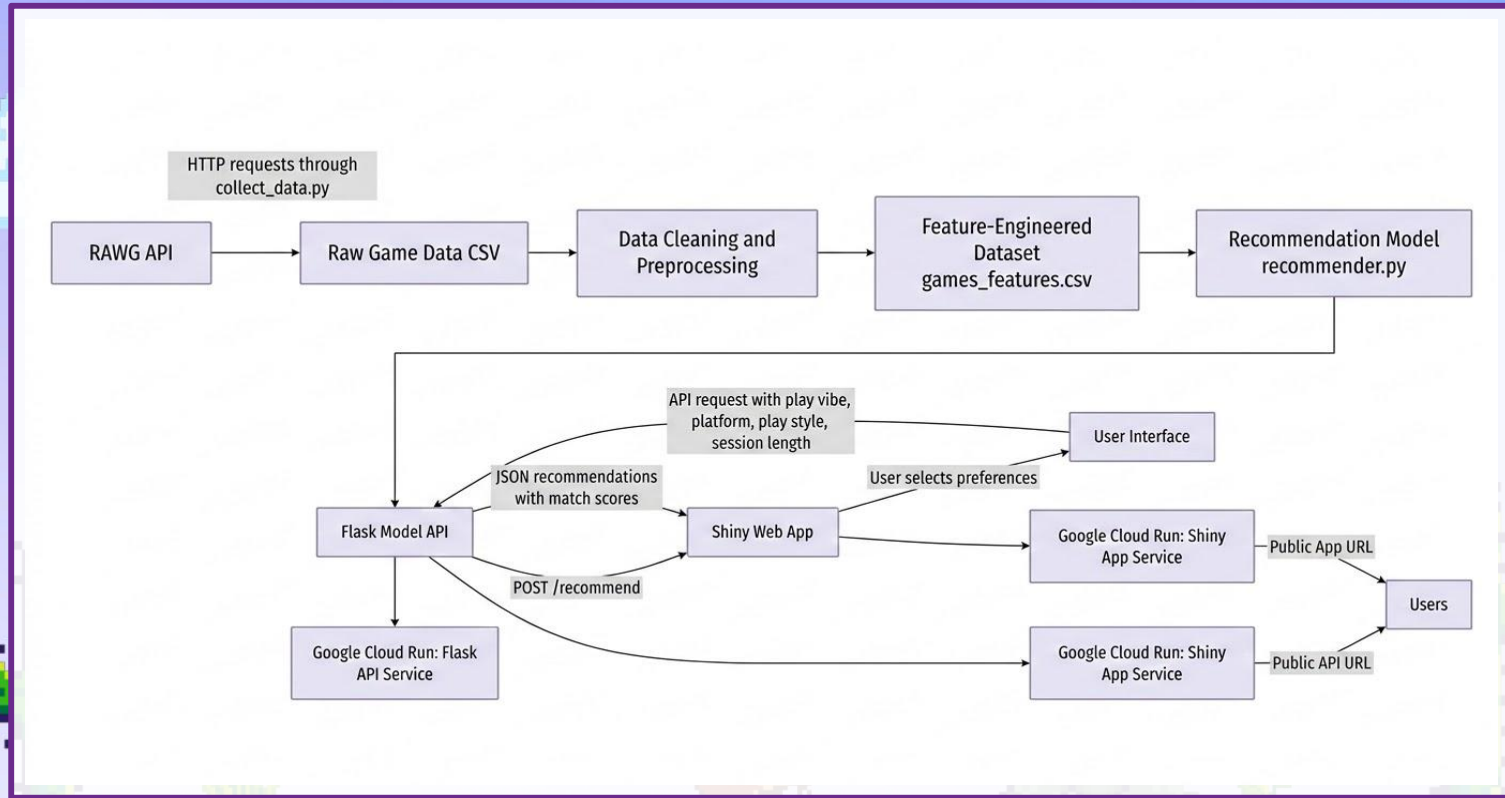
- Play vibe: 45%
- Play style: 20%
- Session length: 15%
- RAWG rating: 15%
- Rating confidence (count): 5%

Evaluation

Manual testing approach

- Tested many input combinations (vibe × platform × style × session length)
- Verified whether results matched selected filter criteria
- Reviewed ranking consistency across repeated queries

Final Solution Architecture Diagram



App Features & Example Output

Choose your preferences

Play vibe

Chill / Cozy

Platform

PC

Play style

Solo

Session length

Short Session (≤ 1 H

Recommendations

3 5 10

Find games

<

Chill / Cozy

Competitive / Intense

✓ Story-Driven

Puzzle / Strategy

Creative / Exploration

✓ PC

PlayStation

Xbox

Nintendo Switch

✓ Solo

Multiplayer / Co-op

✓ Short Session (≤ 1 Hour)

Medium Session (1–3 Hours)

Long Session (3+ Hours)

Found 5 recommendations

Using Chill / Cozy, PC, Solo, Short Session (≤ 1 Hour).

95.7% match

Sonic 3 and Knuckles

Vibe: Chill / Cozy | Platform: PC | Style: Solo, Multiplayer / Co-op | Session: Short Session (≤ 1 Hour)

Rating: 4.41



95.1% match

HoloCure - Save the Fans!

Vibe: Chill / Cozy | Platform: PC | Style: Solo | Session: Short Session (≤ 1 Hour)

Rating: 4.62



Challenges, Solutions & Lessons Learned

Challenges → Solutions

- **Missing playtime data → Treat zero playtime as unknown. Session length was built from genres, tags & playtime.**
- **No direct “play vibe” field in API → Built custom mappings from genres, tags & related metadata.**
- **Too many games scored very high → Made platform a hard filter, enforced one primary vibe per game.**

Lessons Learned

- **Good feature engineering can make a simple recommendation model effective.**
- **A model isn't a product until it has a frontend, an API, and something a real user can actually click.**
- **AI accelerates the work, but the meaningful decisions still required human judgment.**